

Supplementary Information

OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability

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1 Supplementary Note 1: Extended Limitations and Future Work

Top-1 expert purity is modest. The highest-purity atomic expert we identified (E54, English function-words at layer 12) reaches 80%, but most “specialty” experts land at 15–35% purity. Our claims about task-specific experts should be understood distributionally rather than as hard assignments.

Single model family. All results are from Gemma-4-26B-A4B-Instruct. Whether the 96%/4% CPT/SFT decomposition generalizes to Mixtral, DeepSeek-MoE, or GPT-class bio models is an open question. The architectures differ in expert count, top- k selection, and router objective.

Remote-homology ceiling. Our 82.60% remote-homology accuracy leaves an 18% error rate. Specialist methods with structural-alignment heads (ESM-2 + TM-Vec, PLMSearch, CATHe) report 65–75% on differently constructed splits. We hypothesize the remaining gap is a representation-level constraint rather than an SFT-scale constraint; testing requires controlled experiments with more CPT on distant-protein pairs (≈ 150 GPU-hours, deferred).

No chain-of-thought reasoning. We briefly attempted to train with <THOUGHT> reasoning blocks for remote homology but abandoned it because synthetic-CoT templates from sequence features were of insufficient quality. Human-expert or strong-teacher-generated CoT data would likely improve remote-homology performance substantially.

Evaluation scale. BixBench’s True/False subset contains only ≈ 200 questions; small absolute differences in the 90–95% range are within the dataset’s noise floor. Richer evaluation pools (HELM-Bio, LM-BIO) would strengthen the claims.

No ablation on router.proj LoRA. We state that including `router.proj` in the LoRA target modules is critical to enabling expert re-specialization, based on an informal pilot observation. A controlled ablation — CPT with and without `router.proj` LoRA, holding all other hyperparameters fixed — would quantify this claim.

Future experiments. (1) Controlled CPT ablation with richer distant-protein corpora. (2) Router.proj ablation. (3) Human-expert CoT for remote homology. (4) Replication on Mixtral-8x7B or DeepSeek-MoE-16B. (5) Downstream wet-lab validation (variant effect prediction on ClinVar). (6) GGUF quantization for single-GPU deployment on RTX 4090.

2 Supplementary Tables

2.1 Supplementary Table S1: Full Per-Layer JS Divergence

Table 1: Layer-by-layer Jensen–Shannon divergence across training stages. Values are mean pairwise JS across all 28 task pairs at each layer.

Layer	Baseline	CPT	v3 (CPT+SFT)	Δ CPT
0	0.160	0.164	0.164	+0.004
1	0.136	0.113	0.113	−0.023

Table 1 – *Continued*

Layer	Baseline	CPT	v3 (CPT+SFT)	Δ CPT
2	0.172	0.164	0.164	−0.008
3	0.151	0.134	0.134	−0.017
4	0.119	0.113	0.113	−0.006
5	0.159	0.160	0.160	+0.001
6	0.131	0.111	0.111	−0.020
7	0.129	0.123	0.123	−0.006
8	0.134	0.126	0.126	−0.008
9	0.174	0.153	0.153	−0.021
10	0.197	0.159	0.159	−0.038
11	0.160	0.143	0.143	−0.017
12	0.182	0.149	0.149	−0.033
13	0.143	0.127	0.127	−0.016
14	0.157	0.137	0.137	−0.020
15	0.198	0.170	0.170	−0.028
16	0.139	0.121	0.121	−0.018
17	0.184	0.166	0.166	−0.018
18	0.148	0.134	0.134	−0.014
19	0.124	0.123	0.123	−0.001
20	0.140	0.130	0.130	−0.010
21	0.136	0.119	0.119	−0.017
22	0.162	0.149	0.149	−0.013
23	0.157	0.138	0.138	−0.019
24	0.141	0.122	0.122	−0.019
25	0.180	0.149	0.149	−0.031
26	0.161	0.152	0.152	−0.009
27	0.178	0.172	0.172	−0.006
28	0.217	0.197	0.197	−0.020
29	0.240	0.222	0.222	−0.018

61 2.2 Supplementary Table S2: Bootstrap Confidence Intervals

Table 2: Bootstrap 95% confidence intervals for routing metrics (1,000 prompt-level iterations).

Metric	Mean	95% CI Lower	95% CI Upper
Baseline JS	0.344	0.344	0.344
CPT JS	0.440	0.440	0.441
v3 JS	0.444	0.444	0.444
Δ CPT	0.096	0.096	0.097
Δ SFT	0.004	0.003	0.004

2.3 Supplementary Table S3: Format-Matched Routing Control

Table 3: Layer-averaged JS divergence before and after format matching. Six tasks wrapped in a neutral template (### Task: {name} / ### Content: {text} / ### End).

Condition	JS Divergence	Retention Ratio
Original (heterogeneous formats)	0.160	100%
Format-matched (neutral template)	0.145	90.2%

2.4 Supplementary Table S4: Head-to-Head Baseline Comparison (Full)

Table 4: Remote homology accuracy on identical 500-pair balanced evaluation (protein_pair_remote, seed 42). All methods evaluated on the same pair set.

Method	Type	Acc (default)	Acc (optimal)	ROC AUC
ESM-2 650M	PLM (mean-cosine)	50.50%	51.80%	0.513
ESM-2 3B	PLM (mean-cosine)	50.00%	51.20%	0.512
MMseqs2 (easy-search, $s=7.5$)	alignment	54.40%	55.60%	0.556
DIAMOND (more-sensitive)	alignment	53.20%	53.20%	0.532
Gemma-4-Instruct (vocab-ext, ZS)	LM (chat)	60.00%	—	—
OmniGene-4 v3	LM (chat)	59.50%	—	—
OmniGene-4 v4	LM (chat)	82.00%	—	—
OmniGene-4 v5	LM (dual-head)	82.60%	—	—

2.5 Supplementary Table S5: Multi-task Generation Evaluation (v5)

Table 5: Multi-task generation accuracy across six evaluation categories, 100 samples each, on a held-out split. Keyword-overlap is the standard text-overlap score; Structure additionally reports character-level overlap on the 3Di/DSSP output strings.

Category	Keyword score	Char overlap
Protein	1.000	—
Mol	0.980	—
Cell	0.966	—
Literature	0.559	—
Mutation	0.113	—
Structure (gen mode)	0.000	0.259

2.6 Supplementary Table S6: Per-residue Classification Head Accuracy (v5)

Table 6: Per-residue accuracy of the two classification heads on a 50-protein held-out evaluation, under teacher-forced reference-residue inference.

Head	Classes	Chance	Accuracy	N proteins
3Di (Foldseek)	20	5.0%	78.6%	19
DSSP (secondary str.)	8	12.5%	100.0%	31

2.7 Supplementary Table S7: SFT Hyperparameters Across All Stages

Table 7: SFT hyperparameters. v2/v3 used a chat-style template; v4 switched to pure Alpaca with loss masking and task oversampling. v5 adds dual-head classifiers trained jointly with generation.

Parameter	v2 / v3	v4	v5
Init from	v2 / Gemma-bio	v3 LoRA + embed	v4 LoRA + embed
Training data	179K / 199K instr	199K (resampled)	~21K Structure
Prompt template	chat tags	pure Alpaca	pure Alpaca
Loss masking	no	yes (−100)	yes (−100)
Joint loss	gen CE only	gen CE only	0.5 gen + 0.5 cls
Effective batch	64	64	32
Learning rate	5e-5 / 2e-5	2e-5	1e-5
Sequence length	1024	1536	1536
Steps	3,118 / 3,118	4,257	1,350
GPU-hours	11.8 / 13.2	~30	~5

2.8 Supplementary Table S8: Case-Study Pair Details

Table 8: Five case-study protein pairs. ESM2 columns: binary prediction at cosine threshold 0.5. MMseqs/-DIAMOND: 1 if hit found at $E \leq 10^{-3}$, 0 otherwise. v5: parsed answer. Y = correct, N = wrong.

Case	Type	Len1	Len2	Label	ESM-650	ESM-3B	MMseqs	DIAM	v5
1	A	107	106	1	1(Y)	1(Y)	0(N)	0(N)	1(Y)
2	A	142	161	1	1(Y)	1(Y)	0(N)	0(N)	1(Y)
3	B	244	157	0	1(N)	1(N)	0(Y)	0(Y)	0(Y)
4	C	143	101	1	1(Y)	1(Y)	1(Y)	1(Y)	1(Y)
5	D	108	176	1	1(Y)	1(Y)	0(N)	0(N)	0(N)

2.9 Supplementary Table S9: Layer-12 Top-3 Routed Experts per Case

Table 9: Top-3 routed experts at layer 12 for each case-study pair. The same expert set $\{E_{50}, E_9, E_{108}\}$ activates on all five cases, supporting the conclusion that homology decisions occur inside expert computation rather than at the routing gate.

Case	Outcome	Top-1	Top-2	Top-3
1	v5 correct	E50	E9	E39
2	v5 correct	E50	E9	E108
3	v5 correct	E50	E9	E108
4	v5 correct	E50	E108	E9
5	v5 wrong	E50	E108	E9

Layer-12 entropy is stable across cases ($H = 2.70\text{--}2.83$), and per-layer JS divergence between any homologous case and the non-homologous Case 3 stays below 0.04 across all 30 layers (peak at layer 5, 0.040). For comparison, the layer-averaged cross-task JS is 0.230 (Table S1). The $>5\times$ difference supports the gate-vs-expert interpretation in the main text.

2.10 Supplementary Table S10: Per-Task Entropy Analysis

Table 10: Shannon entropy of expert distributions at layer 12 across training stages. Lower entropy = more concentrated expert use.

Task	Baseline	CPT	v3 (CPT+SFT)
NL	3.85	4.13	4.15
DNA	2.89	2.76	2.78
Protein	2.76	2.54	2.56
StdHomology	2.68	2.48	2.50
RemHomology	2.71	2.52	2.54
Cell	2.95	2.88	2.90
Mol	3.02	2.95	2.97
Structure	2.82	2.65	2.67

2.11 Supplementary Table S11: Token-Level Expert Specialization at Layer 12

Table 11: Top-5 tokens for selected experts at layer 12 (v3 model). Purity = fraction of that expert’s activations coming from the dominant modality.

Expert	Purity	Top-5 Tokens
E54	80%	the, of, and, to, in
E59	46%	AA, AT, TT, GG, CC
E78	35%	M, L, V, I, F
E97	28%	CD4, CD8, IL, TNF, IFN
E100	15%	(,), =, C, 0

3 Supplementary Figures

3.1 Supplementary Figure S1: Per-Layer JS Curves

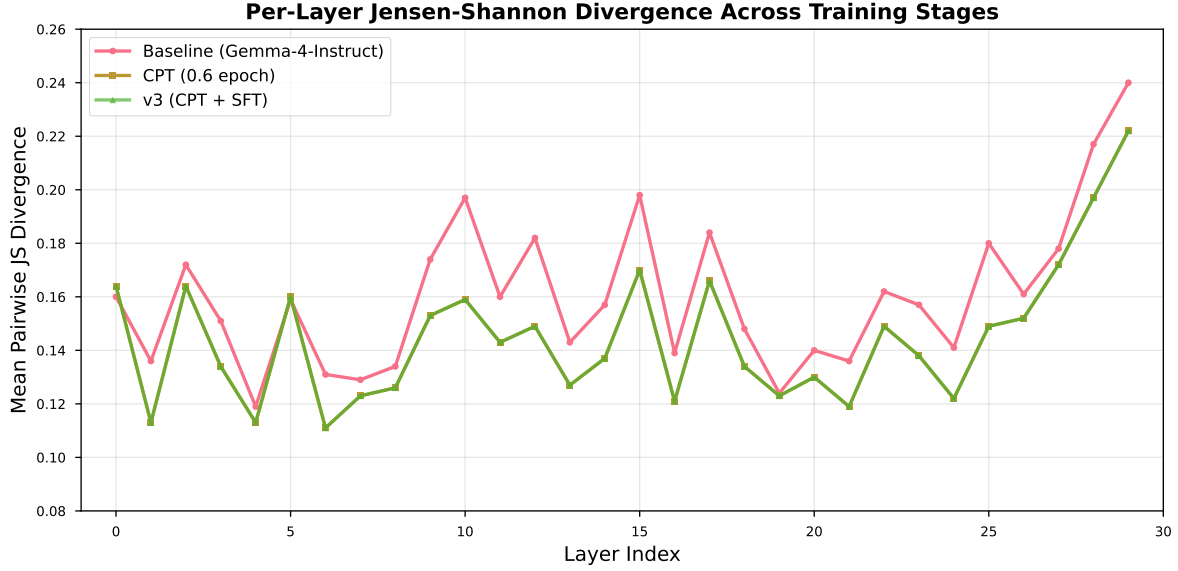


Figure 1: Layer-by-layer Jensen–Shannon divergence across training stages. CPT shows the largest increase in middle layers (L_{11} – L_{22}), while SFT shows concentrated changes near the output layers (L_{28} – L_{29}).

3.2 Supplementary Figure S2: Expert Activation Heatmap

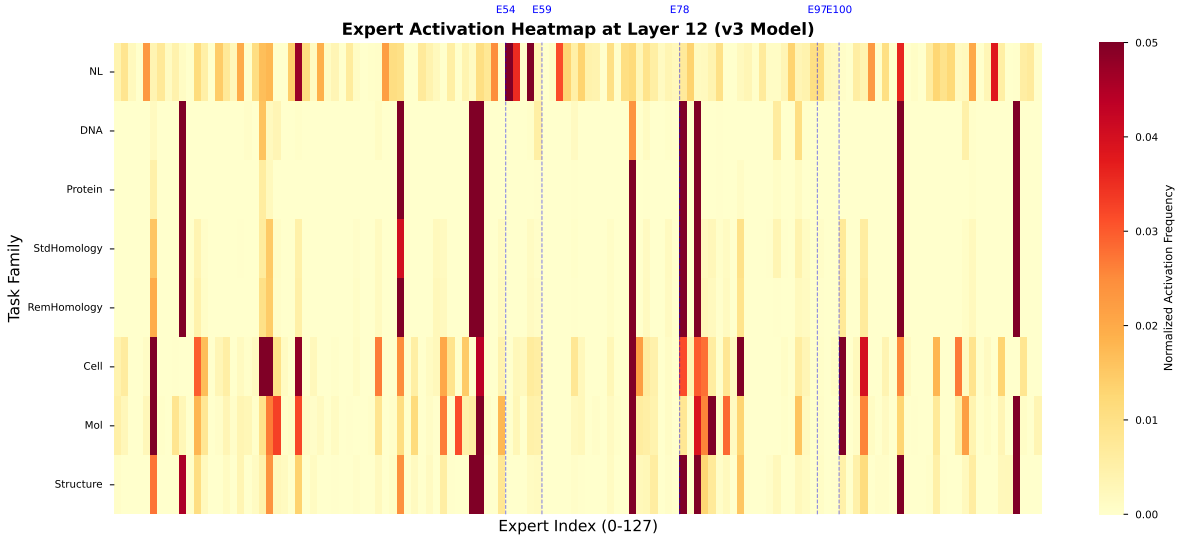


Figure 2: Expert activation heatmap at layer 12 for eight task families. Rows: tasks. Columns: 128 experts. Color intensity indicates normalized activation frequency. Blue dashed lines mark key experts (E54: English function words, E59: DNA 2-mers, E78: amino acids, E97: cell markers, E100: SMILES punctuation).

78 3.3 Supplementary Figure S3: Bootstrap Distribution

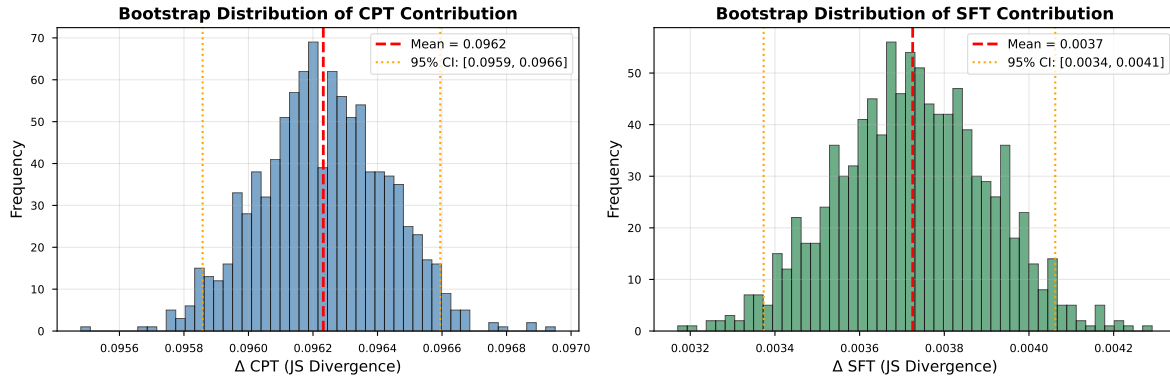


Figure 3: Bootstrap distribution of CPT and SFT contributions to JS divergence (1,000 iterations). Both distributions are narrow and exclude zero. Left: $\Delta \text{CPT} = 0.096$ [0.096, 0.097]. Right: $\Delta \text{SFT} = 0.004$ [0.003, 0.004].

79 3.4 Supplementary Figure S4: Case-Study Routing Mosaic

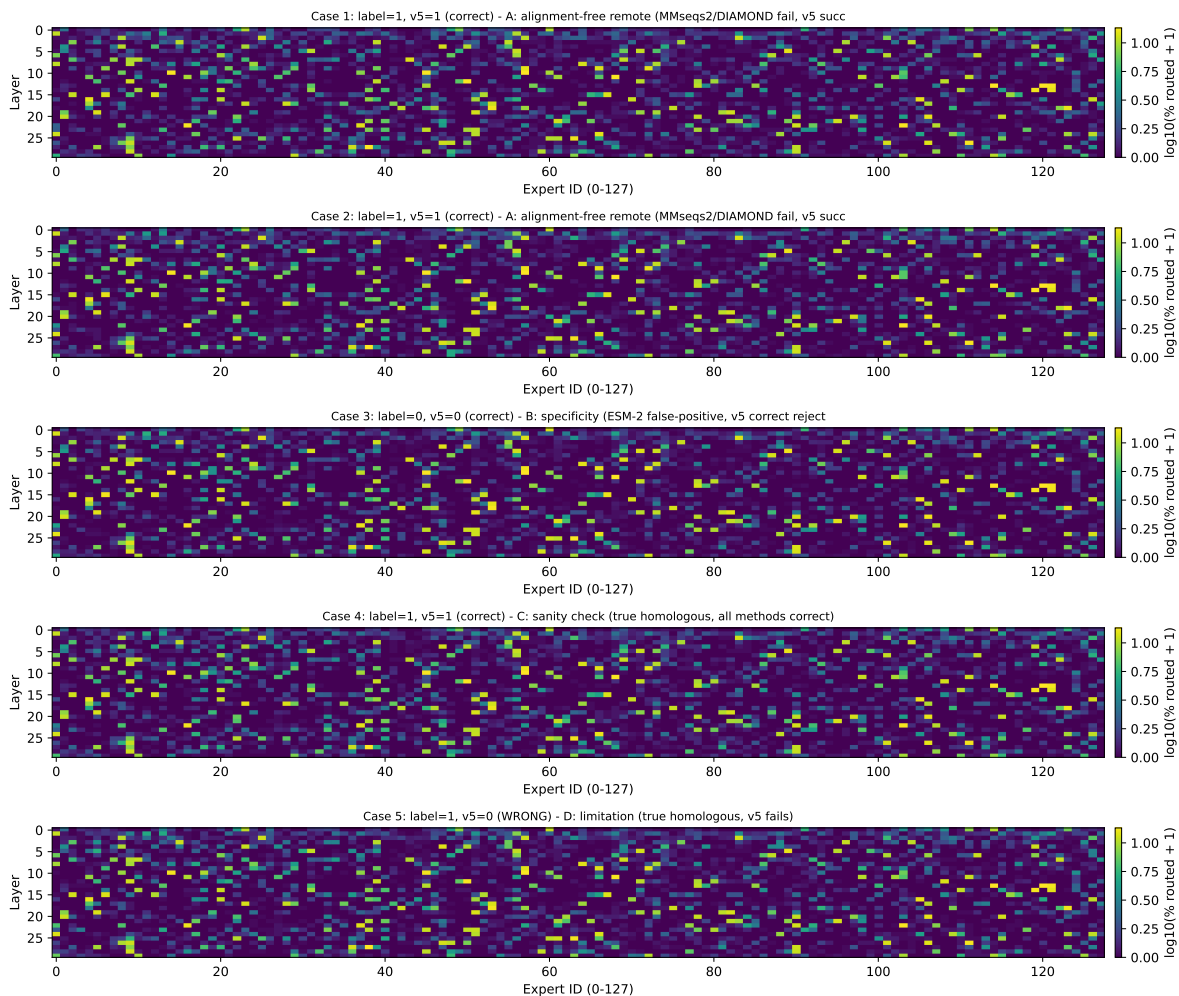


Figure 4: Per-case routing-fraction matrix (30 layers $\times$ 128 experts), log-scaled. Visually similar across all five cases, with the strongest activation cluster near experts 0–20 and 100–115 in middle layers. Consistent with the protein-rich expert population identified in Table S11.

80 4 Code and Model Availability

81 All code, training scripts, and analysis notebooks are available at:

82 <https://github.com/maris205/omnigene4>

83 4.1 Key Scripts

- 84 • biopaws/cpt/1-prepare_cpt_data_mp.py – CPT data preparation
- 85 • biopaws/cpt/2-run_cpt.py – CPT training (8-GPU DDP)
- 86 • biopaws/sft_data/17-train_bio_sft_v3_remote.py – SFT v3
- 87 • biopaws/cpt/40-train_bio_sft_v4.py – SFT v4 (Alpaca + loss-mask)

- `biopaws/cpt/41-train_bio_sft_v5_classifier.py` – SFT v5 (dual-head)
- `biopaws/cpt/43-eval_v5_full.py` – v5 evaluation
- `biopaws/cpt/44-merge_v5_to_full.py` – v5 LoRA → BF16 merge
- `biopaws/cpt/50-esm2_3b_remote_500pair.py` – ESM-2 3B baseline
- `biopaws/cpt/51-classical_baselines_500pair.py` – MMseqs2/DIAMOND
- `biopaws/cpt/52-collect_per_pair_predictions.py` – Per-pair collection
- `biopaws/cpt/53-pick_case_studies.py` – Case-study selection
- `biopaws/cpt/54-extract_routing_heatmap.py` – Router hook extraction
- `biopaws/cpt/55-plot_case_studies.py` – Figure generation

4.2 Model Weights (Hugging Face)

- <https://huggingface.co/dnagpt/OmniGene-4-CPT-v2-merged> – CPT-only, BF16 (~50 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v3-merged> – v3, BF16 (~50 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v3-GGUF> – v3, Q4_K_M (~16 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v4> – v4 LoRA (~1.9 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v5> – v5 LoRA + heads (~1.9 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v5-merged> – v5, BF16 + heads (~52 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-MM-LoRA> – **MM v3 LoRA + extended embedding** (~1.7 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-MM-merged> – **MM v3, BF16, stand-alone** (~49 GB)

5 Supplementary Note 2: OmniGene-4-MM Reproducibility

The multi-modal extension reported in Section 4 of the main text was trained in three sequential LoRA stages on top of the v5-merged checkpoint, on a single H20 GPU, in roughly 1.5 GPU-days total wall-clock. All scripts referenced below live under `omnigene5/scripts/` in the project repository.

5.1 Pipeline Scripts

- `omnigene5/scripts/01-build_unified_jsonl.py` – Assemble a single multi-modal training corpus from Vis-CheBI20, PubMedVision, HPA10M, ChartQA, BiomedVis, and the v5 SFT pool.
- `omnigene5/scripts/02-fix_image_paths.py` – Resolve relative image paths after the corpus assembly step.

- 119 • omnigene5/scripts/30-train_omnigene5_stage1.py – **Stage 1** (vision warmup, $\sim 10\text{K}$
120 steps, LR 5×10^{-5}).
- 121 • omnigene5/scripts/50-train_stage2.py – **Stage 2** (mixed text + vision, 6K steps, LR
122 5×10^{-6}).
- 123 • omnigene5/scripts/82-train_stage3v2.py – Stage 3 v2 (failed: trainable embedding
124 plateaued at 59% standard homology; kept for reproducibility of the failure mode).
- 125 • omnigene5/scripts/83-train_stage3v3.py – **Stage 3 v3 (final)** (LR 2×10^{-5} , frozen
126 embedding, 3000 steps; produces the released checkpoint).
- 127 • omnigene5/scripts/60-eval_stage2.py – Stage 2 evaluation.
- 128 • omnigene5/scripts/91-eval_stage3v2.py – Stage 3 v2 evaluation.
- 129 • omnigene5/scripts/92-eval_stage3v3.py – **Stage 3 v3 evaluation** (BioPAWS standard
130 + remote, Vis-CheBI20 five subtasks, multi-task generation).
- 131 • omnigene5/scripts/95-qualitative_demo_stage3v3.py – Qualitative-showcase generation,
132 39 examples (vision + homology + multi-task), used for Figure 8.
- 133 • omnigene5/scripts/71-router_analysis_mm_extended.py – 8-modality router-level anal-
134 ysis on the Stage 2 checkpoint, used for the modality-invariant-transfer evidence in Section 4.
- 135 • omnigene5/scripts/98-merge_mm_v3.py – Merge MM v3 LoRA + extended embedding into
136 the v5-merged base, producing the stand-alone BF16 checkpoint released as dnagpt/OmniGene-4-MM-merged.

137 5.2 Stage 3 v3 Hyperparameters (Final Checkpoint)

Table 12: Stage 3 v3 training hyperparameters.

Setting	Value
Initialization	v5 LoRA + extended embedding (Stage 2 checkpoint)
Adapter name	stage2 (matches Stage 2’s saved state-dict keys)
LoRA rank / α	$r=64$, $\alpha=128$, dropout 0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj, router.proj
Embedding	Frozen (preserves the v5 paraphrase-transfer signal)
Learning rate	2×10^{-5} (cosine, 3% warmup)
Steps	3,000
Effective batch size	32 (1×32 gradient-accumulation)
Max sequence length	1,536
Optimizer	paged-AdamW-8bit
Precision	BF16
Gradient checkpointing	on (non-reentrant)
Data	45K text (homology oversampled $\times 3$) + 5K vision replay
Hardware	$1 \times \text{NVIDIA H20}$ (80 GB)
Wall-clock	~ 12 hours

5.3 Per-Stage Capability Trajectory

Table 13 reports the evaluation numbers underlying Figure 7. Stage 3 v2 plateaued at the Stage 2 level despite running for 6K steps; freezing the embedding and raising the learning rate $4\times$ in Stage 3 v3 broke the plateau on the first attempt.

Table 13: OmniGene-4-MM capability across all training stages. Vision metrics are accuracy on a 50-prompt-per-task subset of Vis-CheBI20; multi-task generation is keyword-overlap on a 30-prompt-per-category subset of an internal SFT-eval set; homology is BioPAWS standard / remote.

Stage	Vision (Vis-CheBI20)					Multi-task gen				Hom. std/rem
	rec.	cap.	desp	IUPAC	SMILES	Cell	Mol	Prot.	Lit.	
v5 (text only)	—	—	—	—	—	—	—	—	—	99.4 / 82.6
MM Stage 1	1.00	0.90	0.10	0.00	0.00	—	—	—	—	—*
MM Stage 2	1.00	0.88	0.14	0.00	0.00	0.25	0.32	1.00	0.17	59.0 / 56.5
MM Stage 3 v2	1.00	0.90	0.16	0.00	0.00	0.75	0.79	1.00	0.24	59.5 / 60.0
MM Stage 3 v3	1.00	0.96	0.14	0.00	0.00	0.95	0.91	1.00	0.30	85.0 / 69.5

* Stage 1 produces parseable BioPAWS responses on only 4/200 prompts (catastrophic forgetting), so a homology number is not meaningful.

5.4 8-Modality Routing Analysis

The 8-modality routing-divergence numbers reported in the main text are collected by installing `forward_hooks` on all 30 `Gemma4TextRouter` modules of the Stage 2 checkpoint, running 50 prompts per modality across the eight modality categories (*vision_molecule*, *vision_medical*, *vision_pathology*, *vision_chart*, *protein_sequence*, *dna_sequence*, *structure_3di*, *natural_language*), and computing pairwise Jensen–Shannon divergence per layer over the resulting 30×128 expert-count matrices, then averaging over layers. The full 8×8 divergence matrix and per-layer curves are saved to `outputs/OmniGene-4-MM-stage2/router_analysis_8mod/` when the analysis script is re-run; key numbers are reproduced below.

Table 14: Selected pairwise routing JS divergences from the 8-modality analysis (layer-averaged, Stage 2 checkpoint).

Pair	Layer-averaged JS
<i>Within sequence cluster</i>	
protein_sequence vs structure_3di	0.016
protein_sequence vs dna_sequence	0.038
<i>Within vision cluster</i>	
vision_medical vs vision_pathology	0.048
vision_molecule vs vision_chart	0.072
<i>Cross-cluster (representative)</i>	
vision_pathology vs natural_language	1.21
vision_molecule vs protein_sequence	1.24
dna_sequence vs natural_language	1.31

The within-cluster pairs all sit below 0.08, while cross-cluster pairs all exceed 1.2 – a clean three-tier separation that emerges with no modality supervision (the only training signal is token-level

153 cross-entropy). This is the empirical evidence supporting the modality-invariant-transfer claim in
154 the main text.